CreditPathAI

Automating and optimizing the loan recovery lifecycle by modeling repayment behavior using diverse data

Infosys Springboard Internship

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Abstract

- CreditPathAI is an AI-driven loan recovery decision support system.
- Predicts borrower default risk using ML models.
- Classifies borrowers into risk categories (low, medium, high).
- Provides actionable recovery recommendations via dashboard.
- Objective: Reduce NPAs, improve recovery efficiency, support agents.
- CreditPathAI is designed to help banks and NBFCs automate loan recovery decisions. It uses machine learning to predict which borrowers are likely to default and recommends actions based on risk levels. This helps reduce non-performing assets and supports recovery agents with data-driven insights.

Introduction

- BFSI sector faces rising NPAs and inefficient manual recovery.
- Loan defaults lead to financial instability.
- Traditional recovery is costly, time-consuming, and error-prone.
- AI enables automation of risk scoring and recovery prioritization.
- CreditPathAI bridges this gap with an end-to-end ML lifecycle.
- The financial sector is under pressure due to increasing loan defaults. Manual recovery processes are not scalable. CreditPathAI introduces automation using AI to streamline recovery and improve decision making.

Literature Survey

- Credit risk modeling evolved from statistical to ML-based methods.
- BFSI uses ML for fraud detection, scoring, and profiling.
- Kaggle Home Credit dataset provides real-world borrower data.
- Widely used for benchmarking credit risk prediction models.
- Research shows that machine learning models outperform traditional methods in credit risk prediction. The Kaggle dataset used in this project is a standard benchmark in the field, making it ideal for experimentation and validation.

Technologies Used

- Python, Pandas, NumPy – Data processing and analysis.
- Scikit-learn, XGBoost – Machine learning models.
- FastAPI – Backend API for model inference.
- React.js, Plotly.js – Interactive frontend dashboard.
- SQLite – Efficient data storage and joins during EDA.
- The project uses a modern tech stack. Python and its libraries handle data and modeling. FastAPI serves predictions, while React and Plotly power the frontend. SQLite is used for efficient data handling during feature engineering.

Dataset Description

- Source: Kaggle Home Credit Default Risk dataset.
- application_train.csv – Main customer data with target variable.
- bureau.csv – External credit history.
- previous_application.csv – Past loan applications.
- installments_payments.csv, credit_card_balance.csv – Payment behavior.
- The dataset includes multiple files capturing borrower demographics, credit history, and repayment behavior. This comprehensive data enables robust feature engineering and accurate risk prediction.

Methodology

- Data Ingestion – Load and clean raw datasets.
- EDA – Analyze distributions, missing values, correlations.
- Feature Engineering – Aggregate features using SQLite.
- Model Training – Logistic Regression, Random Forest, XGBoost.
- API – FastAPI for serving predictions.
- Frontend – React.js dashboard for visualization and recommendations.
- The project follows a structured ML lifecycle. After cleaning and exploring the data, features are engineered and models are trained. The final model is deployed via API and integrated into a frontend dashboard.

Unique Contributions

- End-to-end ML pipeline from raw data to frontend.
- SQLite-based feature engineering for memory efficiency.
- Rule-based recommendation engine for recovery actions.
- Production-ready API and dashboard architecture.
- Business-focused KPIs and stakeholder usability.
- CreditPathAI stands out for its full-stack implementation. It combines technical robustness with business usability, making it suitable for real-world deployment in financial institutions.

Results & Analysis

- Logistic Regression: AUC 0.74 (baseline).
- Random Forest: AUC 0.73.
- XGBoost: AUC 0.73 with better recall for defaulters.
- Risk Categories: Low (Normal), Medium (Reminder), High (Immediate).
- KPIs achieved: Improved recovery rate, reduced follow-up time.
- The models were evaluated using AUC-ROC. XGBoost performed best in identifying defaulters. The recommendation engine categorizes borrowers and suggests actions, helping agents prioritize efforts.

Conclusion

- CreditPathAI provides actionable insights for recovery agents.
- Enhances risk prediction accuracy and reduces NPAs.
- Improves operational efficiency in loan recovery.
- Demonstrates real-world application of AI in BFSI.
- The project successfully demonstrates how AI can be applied to solve real-world problems in the BFSI sector. It supports agents, reduces defaults, and improves recovery outcomes.

Future Scope

- Integrate LightGBM for improved performance.
- Live deployment with real-time scoring.
- Add authentication and role-based dashboards.
- Continuous monitoring for model drift and retraining.
- Expand to other financial products and geographies.
- Future enhancements include using LightGBM, deploying the system live, adding user roles, and expanding the solution to other financial domains.

THANK YOU